

Algebra II | Guided notes — we fill these in together

Vocabulary

Parent Function: The simplest function in a family; for quadratics this is $y = x^2$.

Reflection: A flip of the graph over an axis; a negative value of a reflects $y = x^2$ over the x-axis.

Vertical Stretch/Compression: When $|a| > 1$ the graph becomes narrower; when $0 < |a| < 1$ the graph becomes wider.

We do together. Describe the transformations from $y = x^2$ to $y = -2(x + 1)^2 + 3$.

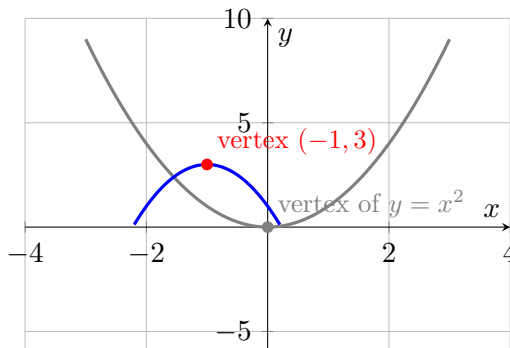
Step 1. The term $(x + 1)$ shifts the graph left by 1 unit.

Step 2. The negative sign in front of the 2 causes a reflection over the x-axis.

Step 3. Since $|a| = 2 > 1$, the graph becomes narrower than $y = x^2$.

Step 4. The $+3$ shifts the graph up by 3 units.

Step 5. Final vertex: $(-1, 3)$.



$y = x^2$ (gray) transformed into $y = -2(x + 1)^2 + 3$ (blue).

You Try. Solve each. Show every step, then name the answer.

- a) Describe the transformations from $y = x^2$ to $y = (x - 2)^2 - 5$. Shift right 2, shift down 5; no reflection or stretch since $a = 1$; vertex $(2, -5)$.

Summary (fill in): Starting from the parent function $y = x^2$, the value h shifts the graph left or right, k shifts it up or down, a negative a causes a reflection over the x-axis, and $|a| > 1$ makes the graph narrower while $0 < |a| < 1$ makes it wider.